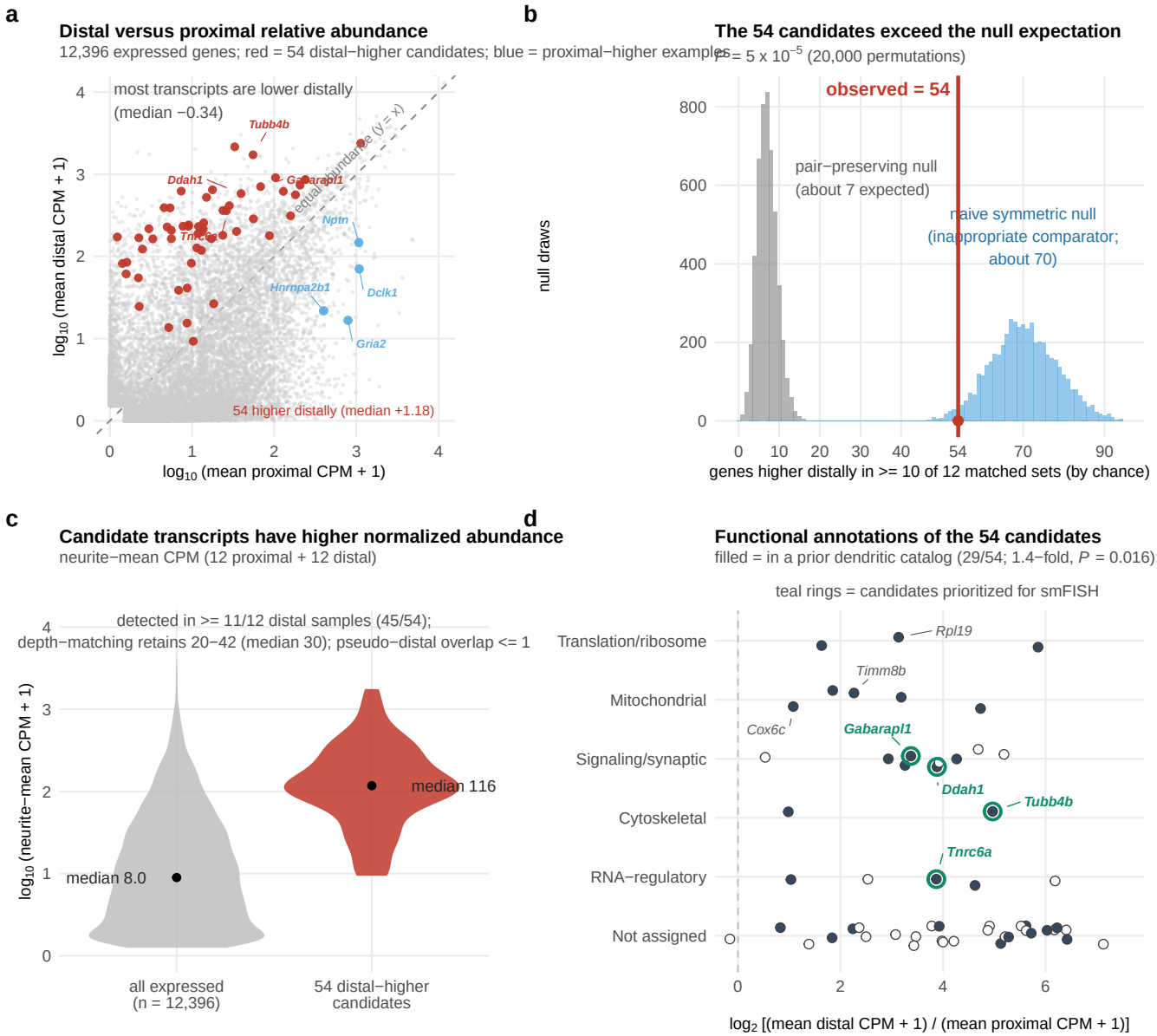

a
Distal versus proximal relative abundance
12,396 expressed genes; red = 54 distal−higher candidates; blue = proximal−higher examples
most transcripts are lower distally
(median −0.34)
Tubb4b
Ddah1
Gabarapl1
Tnrc6a
Nptn
Hnrnpa2b1
Dclk1
Gria2
equal abundance (y = x)
54 higher distally (median +1.18)
log10 (mean distal CPM + 1)
log10 (mean proximal CPM + 1)
b
The 54 candidates exceed the null expectation
P = 5 x 10−5 (20,000 permutations)
observed = 54
pair−preserving null
(about 7 expected)
naive symmetric null
(inappropriate comparator;
about 70)
null draws
genes higher distally in >= 10 of 12 matched sets (by chance)
c
Candidate transcripts have higher normalized abundance
neurite−mean CPM (12 proximal + 12 distal)
detected in >= 11/12 distal samples (45/54);
depth−matching retains 20−42 (median 30); pseudo−distal overlap <= 1
median 8.0
median 116
log10 (neurite−mean CPM + 1)
all expressed
(n = 12,396)
54 distal−higher
candidates
d
Functional annotations of the 54 candidates
filled = in a prior dendritic catalog (29/54; 1.4−fold, P = 0.016);
teal rings = candidates prioritized for smFISH
Translation/ribosome
Mitochondrial
Signaling/synaptic
Cytoskeletal
RNA−regulatory
Not assigned
Rpl19
Timm8b
Cox6c
Gabarapl1
Ddah1
Tubb4b
Tnrc6a
log2 [(mean distal CPM + 1) / (mean proximal CPM + 1)]